Comparing Bayesian Additive Regression Trees, Random Forest and Gradient Boosted Regression ability to Predict Nitrate, Phosphorus and Total Suspended Sediment Event Mean Concentrations in Urban Runoff

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Abstract

Urbanization has led to increases in impervious surfaces, altering hydrologic processes and contributing to increased levels of nutrients and sediment in stormwater runoff. Accurate prediction of Event Mean Concentrations (EMCs) is crucial for developing effective stormwater management practices and predicting pollutant load in receiving waters. This study evaluates the predictive performance of three machine learning models: Random Forest Regression, Gradient Boosted Regression and Bayesian Additive Regression Trees (BART). Data from the National Stormwater Quality Database was used to train the models and make predictions. The models were trained to predict EMCs for nitrate nitrogen, total phosphorus and total suspended solids.

Findings show that Random Forest Regression performed the best for nitrate nitrogen, with no improvement gained from gradient boosting or BART. For total phosphorus, both Random Fores and Gradient Boosted Regression outperform BART. BART performed the best for total suspended solids. The results show that no single model was optimal for all three targets. Overall, this comparative study shows that machine learning methods can effectively model nutrient and suspended sediment loads in runoff, though the optimal model varies by contaminant.

1 Introduction

Urbanization has led to an increase in impervious surfaces like roads and parking lots. These surfaces change the hydrological conditions by preventing infiltration. This leads to higher volume during storm events and increases nutrient and contaminant loads (Mallin et al., 2009). Urban stormwater drainage systems, retention ponds, wastewater plants and stream systems often have higher levels of contaminants due to the collection of contaminants from vehicles, oil and grease and fertilizers (Bell et al., 2017; Taebi and Droste, 2004). This has resulted in a need for accurate predictive models to estimate nutrients and contaminants in runoff, to make informed decisions in stormwater management practices and regulatory compliance.

Event Mean Concentration (EMC) is commonly used to measure nutrient and contaminant concentrations in urban runoff (Lee, 2000). EMC's provide insight into pollutant loads helping

storm water managers to design and implement Best Management Practices (BMPs) to limit nutrients and contaminants in runoff (Maniquiz et al., 2010).

The skewed data distribution and non-linear relationship in water quality data makes modeling with a linear regression difficult (Harrison et al., 2021). A skewed data distribution occurs when the data is not evenly dispersed about the mean, like a bell curve. With skewed data the curve is lopsided, pulled more to one side. Non-linear relationship occurs when the relationship between two variables does not produce a straight line.

To attempt to address these shortcomings, this comparison study uses a comparative analysis of Random Forest Regression, Gradient Boosted Regression and Bayesian Additive Regression Trees (BART). Random Forest Regression is applied to water quality data for its non-parametric approach which better handles skewed data, multicollinearity, non-linear data and high-dimensional data sets (Behrouz et al., 2022; Harrison et al., 2021).

BART provides a Bayesian approach which can better incorporate uncertainty, combine data from multiple sites and update prior and posterior distributions as more data becomes available (Liu et al., 2021; Perera et al., 2021). Runoff sampling is highly variable due to the inconsistency of flow rates given a storm event and the non-uniform movement of contaminants during a storm event. This study aims to quantitatively compare Random Forest, Gradient Boosted Regression and BART models.

2 Literature Review

2.1 Urbanization and Runoff Management

2.1.1 Effect of Urbanization on stormwater runoff

Increased urbanization has led to difficulties in managing pollution loads in stormwater systems. Urbanization increases impervious surfaces like roads and parking lots. These surfaces prevent water from saturating into the ground resulting in increased runoff during storm events (Mallin et al., 2009). Studies have found that urban streams show higher concentrations of pollutants, often due to the collection of pollutants on impervious surfaces that are washed away during storm events.

Sources of pollution include vehicle emissions, oil and grease and fertilizers (Bell et al., 2017; Taebi and Droste, 2004). The first flush phenomenon is an event unique to urban runoff. Due to impervious surfaces ability to collect and store contaminants, the initial runoff from a storm event leads to significant increases in nutrient and pollution loads in water systems (Bell et al., 2017; Mallin et al., 2009; Taebi and Droste, 2004). The first flush event can contribute to the variability in sampling during a storm event.

Urbanization has led to increased areas of impervious surfaces by increasing runoff during storm events, acting as a collection medium for nutrients and pollutants and altering natural hydrological processes that would allow for the natural filtration of pollutants by soil and vegetation (Bell et al., 2017; Mallin et al., 2009; Taebi and Droste, 2004). When stormwater can collect on soil or vegetation, the soil acts as a natural filter as the stormwater makes it way down to the water table. This natural filtering process helps in managing contaminants in water supplies. Increased impervious surfaces in urban areas remove the natural filtering process resulting in higher levels of contamination in streams, lakes and processing facilities.

2.1.2 Event Mean Concentration

Event Mean Concentration (EMC) is a key tool in characterizing runoff nutrient and contaminant levels. EMC is a flow-weighted average of nutrient and contaminated levels in storm runoff (Lee, 2000).

$$EMC = \frac{\sum(C_i * Q_i)}{\sum Q_i}$$

Equation 1.

C_i is the measured concentration, and Q_i is the flow rate. EMC provides a standardized way to calculate nutrient loads in runoff, which is important in estimating pollution loads entering receiving systems (Maniquiz et al., 2010). Receiving systems can be holding ponds where the water is collected and held allowing sediment and contaminants to settle out before being discharged to a natural water way or a water processing facility. Receiving areas can also include settling tanks that are associated with wastewater treatment facilities. In older urban areas the

stormwater and sewer water often flow into the same system. Predicting and managing nutrients is essential in water treatment.

EMCs are a key component in designing Best Management Practices (BMP's) for stormwater management and treatment. EMCs are essential in designing better systems for the removal of pollutants to meet regulatory requirements (Maniquiz et al., 2010). EMC's allow BMP designers to better understand variability in stormwater runoff, and to assess the efficiency of treatment facilities by comparing the EMC at the facility inlet to the EMC at the outlet of a treatment facility(Lee, 2000; Maniquiz et al., 2010).

2.1.3 Nutrients of Concern and Effects on Ecosystems

Increased nutrients introduced to aquatic environments can have a significant impact on those ecosystems through eutrophication (Rathore et al., 2016). Eutrophication occurs when there is an increase in nutrients that leads to the rapid development of algae blooms, some of which can be toxic to aquatic organisms. In turn, increased algae blooms can lead to decreases in oxygen levels. When the algae blooms die, they are decomposed by bacteria, a process that utilizes oxygen. Decreasing oxygen levels have a negative impact on fish and other aquatic organisms resulting in a loss of biodiversity (Rathore et al., 2016). Eutrophication can negatively affect the processes in holding and settling tanks as well. Drastic increases and decreases in oxygen due to circumstances detailed above can impact a pond or tanks ability to aerobically breakdown contaminants.

Two main nutrients that affect eutrophication in aquatic ecosystems are Nitrogen and Phosphorus. Elevated nitrogen levels can lead to an increased growth of algae blooms leading to eutrophication and potentially toxic algal blooms (Camargo and Alonso, 2006). Certain forms of Nitrogen such as Nitrate, Nitrite and Ammonia can be toxic to aquatic organisms. Ammonia is highly toxic to aquatic organisms, affecting their ability to reproduce (Camargo and Alonso, 2006). Elevated Phosphorus levels, like nitrogen, can also lead to increased algae blooms leading to eutrophication and potentially toxic algae blooms (Yang and Toor, 2018). Phosphorus can also bind to sediment in bodies of water. If the sediment is disturbed, phosphorus is released back into the water. This can perpetuate a eutrophication cycle (Yang and Toor, 2018).

Suspended solids also play a critical role in evaluating runoff and managing healthy aquatic ecosystems. Abnormally high levels of suspended solids can lead to increased turbidity in aquatic

ecosystems. Turbidity reduces the amount of light penetration resulting in diminished growth of aquatic plants which are necessary as a food source and to produce oxygen (Quinteiro et al., 2015). Nutrients and contaminants of concern can bind to suspended solids. Nitrogen, Phosphorus, heavy metals and hydrocarbons can be transported to aquatic habitats. Increases in Nitrogen and Phosphorus may lead to eutrophication and increases in heavy metals and hydrocarbons can lead to elevated toxicity levels in aquatic ecosystems (Quinteiro et al., 2015).

2.2 Predicting Nutrient Concentrations in Urban Runoff

2.2.1 Random Forest Regression

Water quality monitoring programs have several challenges that can be addressed using Random Forest Regression. Water quality data often shows a skewed distribution. Skewed data can be problematic for linear regression analysis. Random Forest is a non-parametric method that is better suited for handling skewed data (Harrison et al., 2021). The relationship between environmental variables and nutrient concentrations is often nonlinear. This is problematic for traditional linear regression models. Random Forest Regression is more capable of modeling nonlinear relationships, leading to more accurate predictions (Harrison et al., 2021). Water quality data often shows multicollinearity between predictor variables. Random Forest Regression can address this issue by using random subsets of predictors for trees. This can reduce the impact of multicollinearity and improve the model's robustness (Harrison et al., 2021; Thompson and Dickenson, 2021).

Random Forest Regression is not prone to overfitting, especially when dealing with high-dimensional data. Random Forest's ability to determine variable importance is also important when dealing with high-dimensional data (Behrouz et al., 2022). The Random Forest Regression model is beneficial in modeling water quality due to its ability to utilize many predictors, determine variable importance and account for multicollinearity and non-linearity in the data (Behrouz et al., 2022; Harrison et al., 2021; Thompson and Dickenson, 2021).

Climatological and catchment characteristics are the most important in predicting even mean concentrations (Behrouz et al., 2022). Significant climatological characteristics included precipitation, duration of rainfall events, intensity of rainfall and antecedent dry period (Behrouz et al., 2022). The antecedent dry period is the period without rainfall prior to a rainfall event.

Significant catchment characteristics included slope, area, imperviousness, saturated hydraulic conductivity and available water capacity (Behrouz et al., 2022).

2.2.2 Gradient Boosted Regression

Gradient boosted regression trees, like random forest regression trees, are beneficial in modeling storm event runoff because of their ability to capture complex, nonlinear relationships in high-dimensional data. The boosting process involves sequentially adding weak learners to improve the models fit. This process can lead to overfitting if the model is not appropriately tuned. Boosted models' emphasis is on reducing bias, which can increase variance if hyperparameters are not properly regulated (Alnahit et al.,). Alnahit et al., found that boosted regression was more prone to overfitting than random forest regression.

2.2.3 Bayesian Additive Regression Tree's

Utilizing a Bayesian approach to modeling EMC could potentially provide several improvements. First, the Bayesian approach allows for the combination of data from different sampling sites as well as incorporating new data. Applying a Bayesian model allows for updates to the prior probability based on the posterior probability and then updating the posterior probability on newly acquired data. When new data is available, the preciously calculated posterior distribution is used as the new prior distribution. This allows for more computational efficiency when new data is acquired by not having to retrain the model on previous data (Gronewold and Borsuk, 2010; Liu et al., 2021).

Second, a Bayesian approach better accounts for the incorporation of uncertainty in the parameter estimation, model selection and prediction uncertainty. It does not provide a single estimate for a parameter but rather a distribution of possible values. This helps to address uncertainty in stormwater predictions (Liu et al., 2021; Perera et al., 2021). Due to their ability to incorporate information regarding covariates and data from multiple sites, Bayesian models tend to have tighter uncertainty bounds (Potash and Steinschneider, 2022).

Bayesian models use uncertainty in model and parameter estimation to estimate predictive uncertainty. A Bayesian model will provide a probability distribution for predicted values. Rather than generating a single prediction, it provides a range of possible values that captures the uncertainty in the measurements and models (Gronewold and Alonso, 2006). This could be

beneficial from a compliance standpoint. By generating a probability distribution for the prediction, water treatment administrators can determine the likelihood of complying for a given runoff event (Liu et al., 2021; Perera et al., 2021).

3 Data Description

3.1 Data Source and Structure

Data sourced from the National storm water quality database (NSQD). The NSQD is an urban stormwater runoff database that collected stormwater data over a fifteen-year period from 2001 - 2015. Over 9,000 events were sampled from 200 municipalities in 9 different EPA rain zones (National Stormwater Quality Database, 2015).

The NSQD is in a tabular format and includes information on key watershed characteristics and laboratory results for several common contaminants of concern in environmental sampling including the projects targets, nitrate nitrogen, total phosphorus and total suspended solids. Catchment characteristics include but are not limited to land use designation, percent impervious, run off depth, latitude and longitude and drainage area in acres. In total, there are 9,052 observations and 259 parameters (National Stormwater Quality Database, 2015).

3.2 Limitations in the data

Several key parameters were not routinely recorded during storm events. End date and time for storm events were missing, making it impossible to calculate storm duration and rate of rainfall. Antecedent dry period, the number of days since the last storm event, was often missing in the data. Behrouz et al. showed these to be important parameters when modeling stormwater runoff events. Several important catchment characteristics were also missing including, available water capacity for the soil in the region, saturated hydraulic conductivity and slope. These catchment characteristics have been shown to be beneficial when modeling nutrient loads in stormwater runoff (Behrouz et al., 2022).

The nature of the sampling resulted in limited options for handling missing data. With data being collected from 700 different sampling sites across the country, missing data was typically consistent for a particular site. Meaning, they typically never recorded the data for a particular

parameter like antecedent dry period while other sites recorded antecedent dry period but not the end date for a rain event. Given the geographic differences between sites, imputing observations for one site based on values on another site was not appropriate. As a result, rows with missing values were dropped, resulting in 884 rows for modeling.

3.3 Data Preprocessing

The NSQD contained laboratory results for several environmental contaminants. The data also included the laboratory quality control (QC) results for each analysis. The QC results are independent of the storm event and not relevant to the model, as such they were dropped. Several results were non-detect(ND), meaning they were below the reporting limit (RL) for that analytical method. These results were typically recorded as 'ND' or '< RL' . These results were converted to zero for use by the models.

Many microbiology methods contained results '> threshold'. The threshold is dependent on the dilution performed at the time of analysis. If the improper dilution was performed, they were unable to accurately quantify the microbe of concern. Unlike the less than results that have a natural lower limit, zero, the greater than results do not have a natural limit. As a result of this lack of limit, the greater than results were dropped.

3.3.1 Encoding

Several categorical variables were included in the dataset, and because these variables do not contain any inherent ordering they were treated as nominal rather than ordinal. To ensure the machine-learning models used could interpret these variables appropriately, they were transformed using the 'OneHotEncoder' function from the scikit-learn library. This created a series of binary indicator variables for each category.

3.3.2 Feature Engineering

Multiple features were engineered from the NSQD data. Several location identifiers were provided in the dataset including latitude and longitude, location code, jurisdiction county and city, and state. The latitude and longitude were kept, and all other location variables were dropped. The latitude and longitude were used to calculate the distance of each sampling site from Omaha, NE using the geopy point and geopy distance functions from the Geopy library.

The runoff coefficient and volumetric runoff coefficient were calculated using data from the NSQD. The runoff coefficient was calculated using the total runoff depth in inches divided by the precipitation depth in inches (Akan, 1993).

$$Runoff\ Coefficient = \frac{Runoff(in)}{Precipitation\ Depth(in)}$$

Equation 2.

The volumetric runoff coefficient was calculated using the percent impervious value provided by the dataset and using the assumed 0.9 impervious factor for impervious surfaces and 0.05 factor for pervious surfaces (Akan, 1993).

$$Calculated\ RV = percen\ impervious * 0.9 * 0.05$$

Equation 3.

3.4 Feature Selection

Three techniques to establish feature importance were utilized in this project. f regression, random forest regression feature importance and recursive feature importance(RFE). All three were implemented using their respective functions in the scikit-learn feature selection package. Each method returned similar sets of features with different orders of importance. The total suspended solids model was trained on all three feature sets. RFE features produced the best performing model. Due to time constraints and limits in computational resources, the nitrate nitrogen and total phosphorus models were only trained on the RFE features. Results for this project were generated using the RFE features.

4 Exploratory Data Analysis (EDA)

4.1 Descriptive Statistics

The exploratory data analysis showed that the targets nitrate nitrogen, total phosphorus and total suspended solids were heavily right skewed and contained many outliers. Nitrate nitrogen has a mean of 0.1426 and a range of 0 – 16.79 (see Table 1). The data is highly skewed to the right with several outlier's present (see figure B1 and B2).

Mean	0.142
Standard Deviation	0.816
Minimum	0.0
Maximum	16.79

Table 1. Descriptive statistics for Nitrate Nitrogen.

Total phosphorus is also highly skewed to the right with several outlier's present (see figures B3 and B4). The mean value of the phosphorus data is 0.401 with a minimum of 0.0 and a maximum value of 10.2 (see table 2).

Mean	0.401
Standard Deviation	0.652
Minimum	0.0
Maximum	10.20

Table 2. Descriptive statistics for Total Phosphorus.

Total suspended solids showed a right skewness as well, though not as severe as for nitrate nitrogen and total phosphorus. The data was slightly distributed along the lower end of the range (see figures B5 and B6). Total suspended solids have a mean of 247.89 and a range from 0.0 – 2490.0 (see table 3).

Mean	156.56
Standard Deviation	247.89
Minimum	0.0
Maximum	2490.0

Table 3. Descriptive statistics for Total Suspended Solids

4.2 Correlation

Correlation heatmap plot for nitrate nitrogen showed strong correlations with total kjeldhal nitrogen, nitrite nitrogen and several heavy metals including chromium, copper, nickel and zinc. Nitrate nitrogen only showed a moderate to weak correlation with catchment characteristics such as drainage area and precipitation depth (see figure B7). Total phosphorus showed a strong correlation with several parameters including nickel, zinc and total solids. Like nitrate nitrogen, total phosphorus showed a week correlation to catchment characteristics such as runoff in inches and calculated RV (see figure B8). Total suspended solids show a strong correlation with several heavy metals including chromium, lead, nickel and zinc. Like the other two targets, total suspended solids only has a weak correlation with catchment characteristics such as percent impervious and calculated curve number (see figure B9).

Exploratory data analysis showed highly skewed distributions for each of the target variables with several outliers for each target. Analyzing the correlation between the target variables and predictive variables showed the targets are more strongly correlated with nutrients and contaminants than they are with the catchment characteristics.

5 Methodology

Random Forest Regression, Gradient Boosted Regression and Bayesian Additive Regression Trees were each used to model and predict the target variables nitrate nitrogen, total phosphorus and total suspended solids. Random Forest Regression had been shown to be capable of successfully modeling nutrient loads in urban runoff (Behrouz et al., 2022). Gradient Boosted Regression was used as a comparison to see if any gains in prediction accuracy could be made on the Random Forest Regression Model. BART was implemented to add a Bayesian component to the modeling. Bayesian modeling has been shown to better capture uncertainty in stormwater runoff modeling (Liu et al., 2021; Perera et al., 2021).

Data was split into train and test sets with an 80/20 ratio using the 'train_test_split' function in the 'model_selection' package of the scikit-learn library. The 'GridSearchCV' and 'KFold' functions, also part of the 'model_selection_package' in scikit-learn, were used to run 5-fold cross validation and hyperparameter tuning for each model and target. The training data was randomly split into five parts. Four training portions and one validation portion. The model is iteratively trained leaving a different portion out as the validation set.

5.1 Random Forest Regression

Random Forest Regression was implemented using the 'RandomForestRegression' function found in the scikit-learn ensemble package. Random forest is an ensemble method based on a decision tree model. A decision tree will look at all available parameters and make a split using the best parameter for that split. This is called a top-down greedy approach. For a regression model, the best split is the one that minimizes the residual sum of squares (James et al., 2013). At each node, a split is made resulting in 2 branches until the terminal node, or leaf is reached. The terminal nodes represent a fixed number of non-overlapping prediction spaces. The value of that leaf is the mean of the responses of the observations that fall there (James et al., 2013).

With random forest regression, rather than training a tree on all available parameters, a randomly selected subset of parameters is used to make the split at each node. The result is many different trees being trained on subsets of the same data. This method of using different subsets decorrelates the trees because each tree is seeing a different subset of parameters. The trees are finally averaged to produce the final model (James et al., 2013).

The 'GridSearchCV' function was used during the training for hyperparameter tuning. The GridSearchCV function takes a matrix of hyperparameters and corresponding possible values and trains a model on each possible combination. The out of bag score is used to determine which set of hyperparameter settings results in the best functioning model.

Hyperparameter	Description
n estimators	Number of trees in the forest.
Max depth	Maximum depth of the tree.
Minimum sample split	Minimum number of samples required to split an internal node.
Minimum sample leaf	Minimum number of samples required to be at a terminal (leaf) node.
Maximum features	Number of features to consider when looking for the best split.
Maximum leaf nodes	Maximum number of terminal (leaf) nodes.
Minimum impurity decrease	Node will be split if the split induces a decrease in impurity greater than or equal to this value.

Table 4. Hyperparameter selection for random forest regression.

Hyperparameter	Nitrate Nitrogen	Total Phosphorus	Total Suspended Solids
n estimators	150	100	100
Max depth	5	15	15
Minimum sample split	2	2	2
Minimum sample leaf	3	2	2
Maximum features	Log2	Log2	None
Maximum leaf nodes	None	None	None
Minimum impurity decrease	0.0	0.0	0.0

Table 5. Hyperparameter values for each random forest regression target.

5.2 Gradient Boosted Regression

Gradient boosted regression was implemented using the 'GradientBosstedRegression' function from the scikit-learn ensemble package. Gradient boosted regression is also a decision tree based ensemble method. Gradient boosted trees are an additive model trained in a sequential manner. At each stage a decision tree is fitted to the negative gradient of the loss function, squared error. Each base learner, or tree, is trying to move the model in the direction of deepest descent. After each tree is fitted, the predictions are multiplied by the learning rate before being added to the existing model. The learning rate is a weight applied to the predictions to slow the rate of learning for each model in the sequence. A lower learning rate forces the model to make incremental adjustments at each stage (scikit-learn, 2019).

'GridSearchCV' was also used to implement 5-fold cross validation and hyperparameter tuning. Many of the hyperparameters for gradient boosted regression are the same as random forest regression, with a few exceptions. Learning rate, which was discussed previously, and loss. Loss defines the method for determining the error at each stage of modeling. In this case squared error was used.

Hyperparameter	Nitrate Nitrogen	Total Phosphorus	Total Suspended Solids
n estimators	100	100	100
Max depth	10	10	10
Minimum sample split	2	2	2
Minimum sample leaf	3	2	2
Maximum features	None	Log2	Log2
Maximum leaf nodes	10	None	None
Minimum impurity decrease	0.05	0.05	0.05
Learning Rate	0.2	0.1	0.1

Table 6. Hyperparameter values for gradient boosted regression.

5.3 Bayesian Additive Regression Trees

Bayesian additive regression trees were implemented in R through the dbarts library. The dbarts package was accessed via the parsnips package in the tidymodels ecosystem. Dbarts constructs an ensemble of regression trees, each contributing to the final model. BART is inherently Bayesian, as such priors are placed on the structure and depth of the trees as well as on the terminal node predictions. Posterior inference is performed using Markov Chain Monte Carlo (MCMC) sampling. This results in a posterior distribution over possible tree ensembles, providing both point predictions and uncertainty intervals (*R: Bayesian Additive Regression Trees*, 2025).

Within the parsnip package hyperparameters such as number of trees and prior terminal node coefficient can be tuned. Other dbarts engine hyperparameters such as burn in, number of chains and number of posterior samples are passed directly to dbarts. Cross validation showed the ideal number of trees for phosphorus and nitrate is 100 and 150 for total suspended solids. The tuned prior terminal node coefficient for phosphorus and nitrate is 0.95 and 0.98 for total suspended solids (*R: Bayesian Additive Regression Trees*, 2025).

6 Results and Evaluation

The model's performances were evaluated using root mean squared error (RMSE). The smaller the RMSE value, the better the model's performance. The best performing model for nitrate nitrogen was random forest regression with an RMSE of 0.13. The clearly outperformed gradient boosted regression and BART with RMSE's of 0.40 and 0.45, respectively

Random Forest Regression	Gradient Boosted Regression	BART
0.13	0.40	0.45

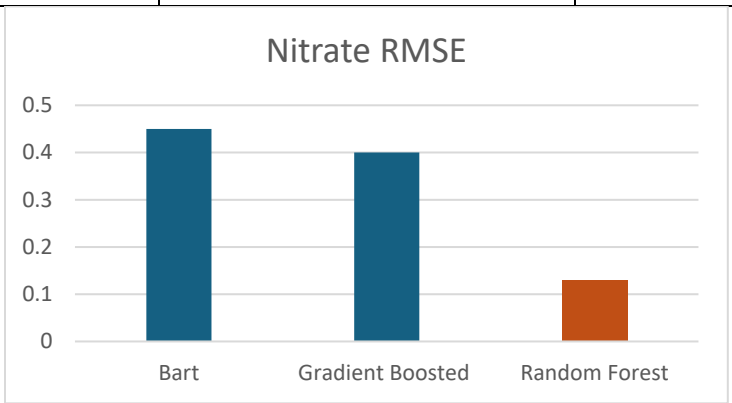

Figure 1. Nitrate nitrogen RMSE results.

The best performing models for total phosphorus were random forest regression and gradient boosted regression. Each with an RMSE of 0.78. Outperforming the BART model with an RMSE of 0.81.

Random Forest Regression	Gradient Boosted Regression	BART
0.78	**0.78**	0.81

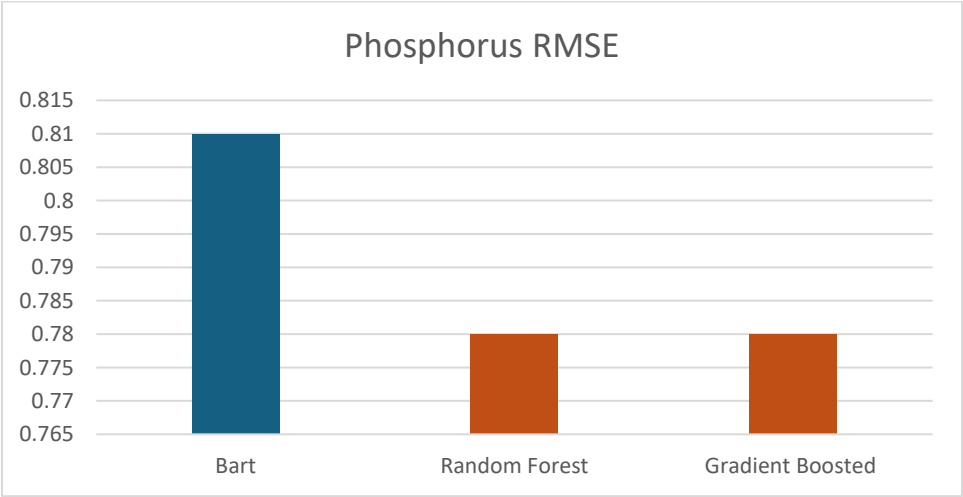

Figure 2. Total phosphorus RMSE results.

For the total suspended solids model, BART was the top performer with an RMSE of 141.685. Random forest regression and gradient boosted regression did not perform as well with RMSE's of 154.28 and 142.33 respectively.

Random Forest Regression	Gradient Boosted Regression	BART
154.28	142.33	**141.685**

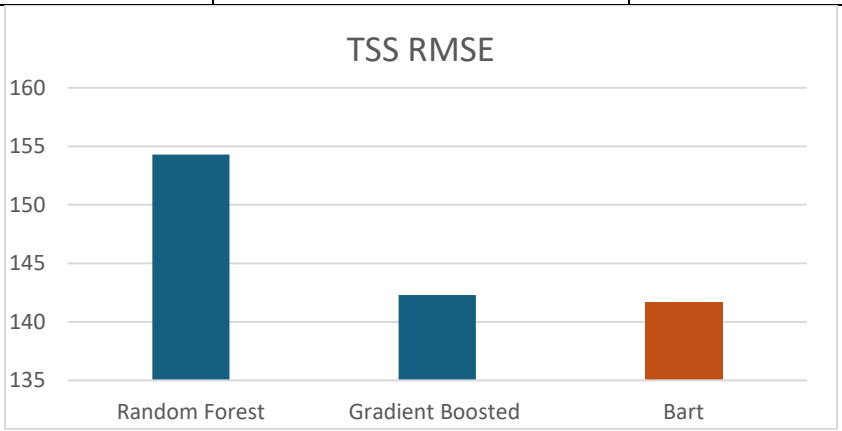

Figure 3. Total suspended solids RMSE results.

Based on the RMSE results I would recommend using random forest regression for nitrate, random forest regression or gradient boosted regression for phosphorus and BART for total suspended solids.

7 Discussion

Overall, the models showed useful predictive capabilities. The strong performance of the tree-based methods shows the problem is well suited to nonlinear modeling approaches. Except for total suspended solids, the Bayesian model did not provide any improvements in performance.

The number of observations with missing values presented a challenge with approximately ten percent of observations being used in the modeling process. The nature of the sampling made imputation difficult to implement and was not attempted as a result. Training three models to predict three different targets was another challenge I encountered. This approach required considerable computational resources and was very time-consuming. This could be addressed in future work by utilizing multiple target model or limiting the scope to a single target.

8 Conclusion

Random forest regression performed the best for nitrate nitrogen concentrations, and neither the gradient boosted regression nor BART model yielded improvements. For total phosphorus, both random forest and gradient boosted regression produced stronger predictive accuracy than the BART model. BART was the top performing model for total suspended solids.

Except for total suspended solids, the Bayesian model did not produce any improvements in model accuracy. Future work could include using a multioutput model to make predictions on multiple targets utilizing one model. A more complete data set would be beneficial in future work. Compiling data for several parameters that were missing including precipitation duration, precipitation intensity, available water capacity and saturated hydraulic conductivity could be beneficial.

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10 Appendices

Appendix A

Links to Code

NSQD Data cleaning

Exploratory Data Analysis

Nitrate Nitrogen - Random Forest and Gradient Boosted Regression

Total Phosphorus - Random Forest and Gradient Boosted Regression

Total Suspended Solids - Random Forest and Gradient Boosted Regression

BART Models for all three targets

Appendix B

Additional Plots

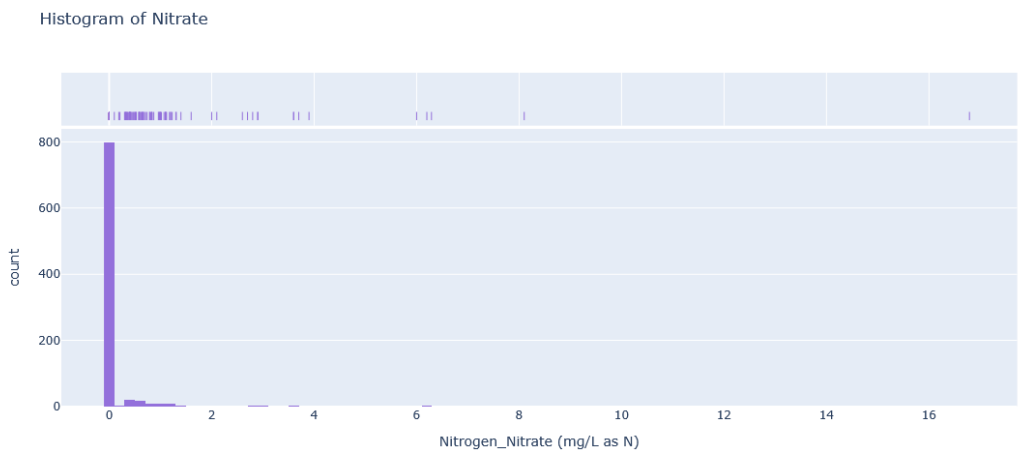

Figure B1. Histogram of Nitrate Nitrogen.

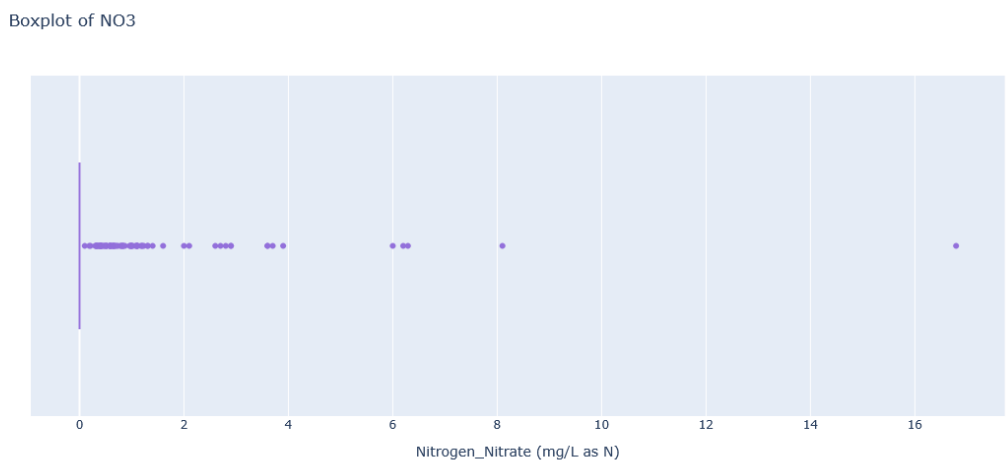

Figure B2. Boxplot of Nitrate Nitrogen.

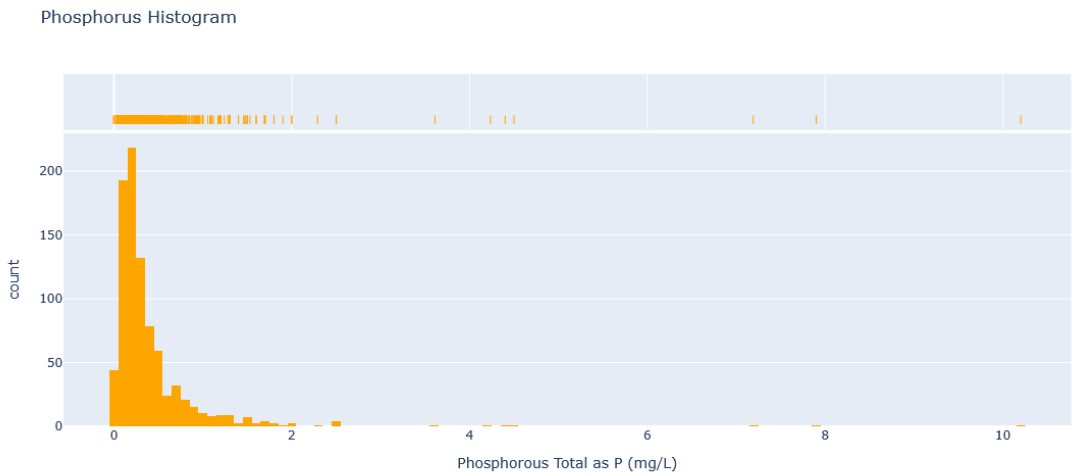

Figure B3. Histogram of Total Phosphorus.

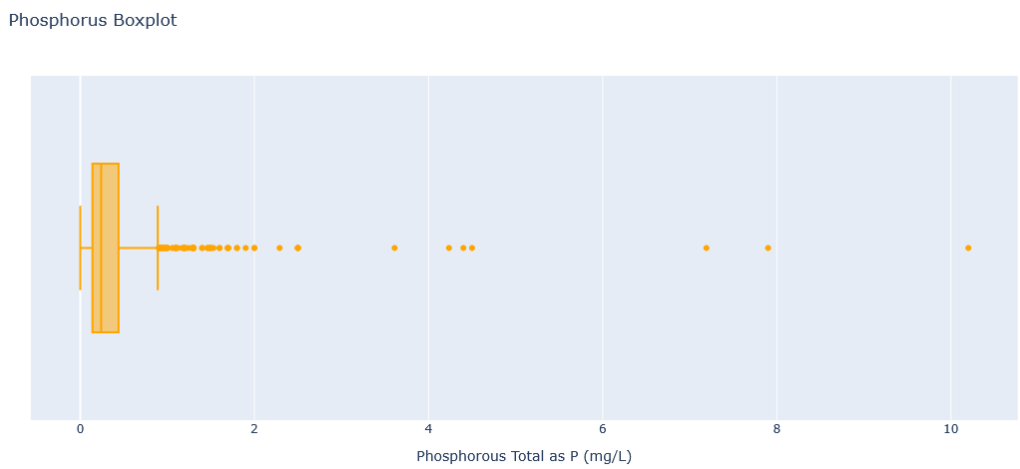

Figure B4. Boxplot of Total Phosphorus

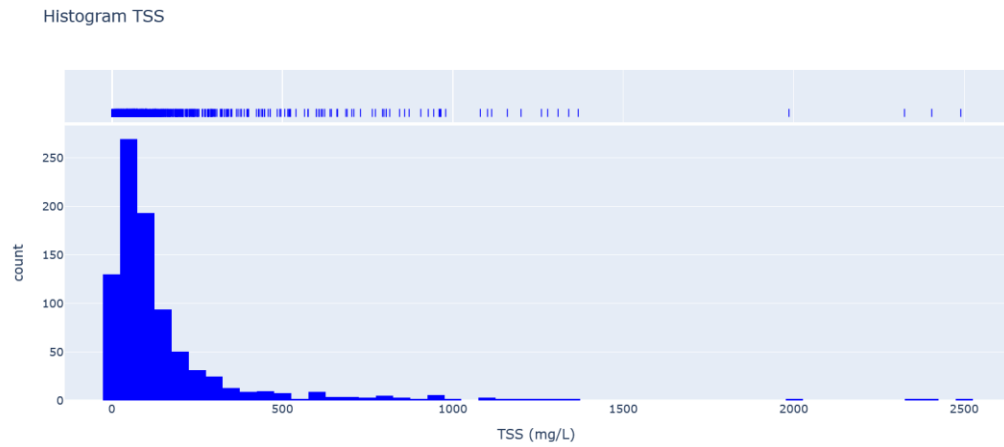

Figure B5. Histogram of Total Suspended Solids.

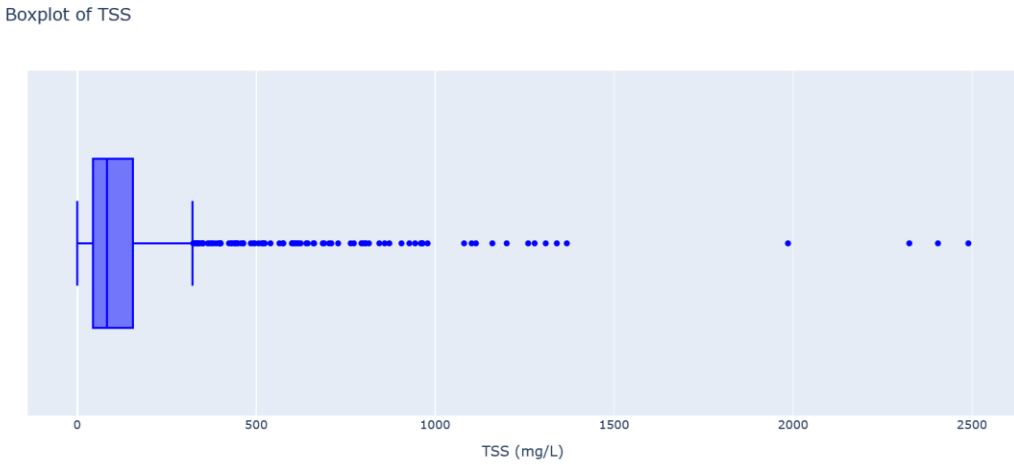

Figure B6. Boxplot of Total Suspended Solids

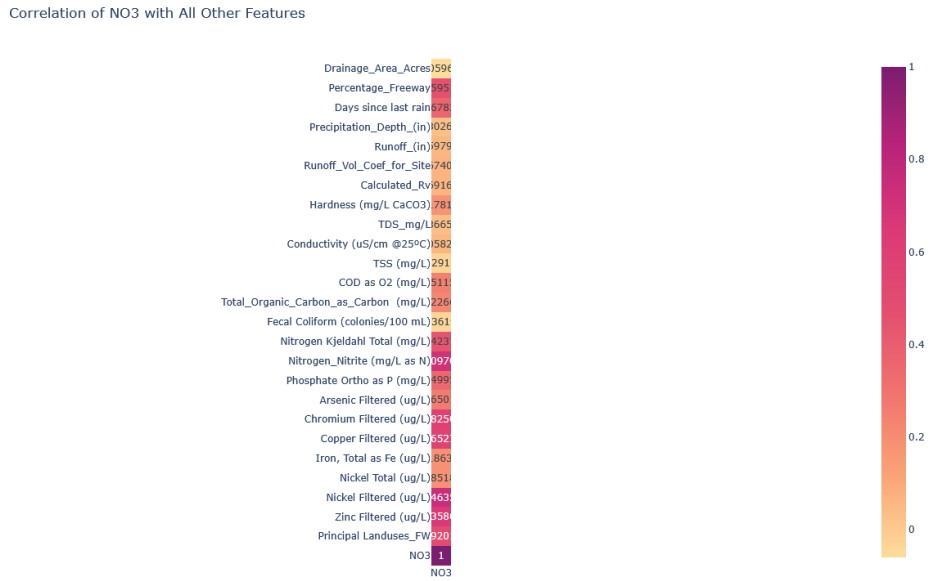

Figure B7. Correlation heatmap for Nitrate Nitrogen

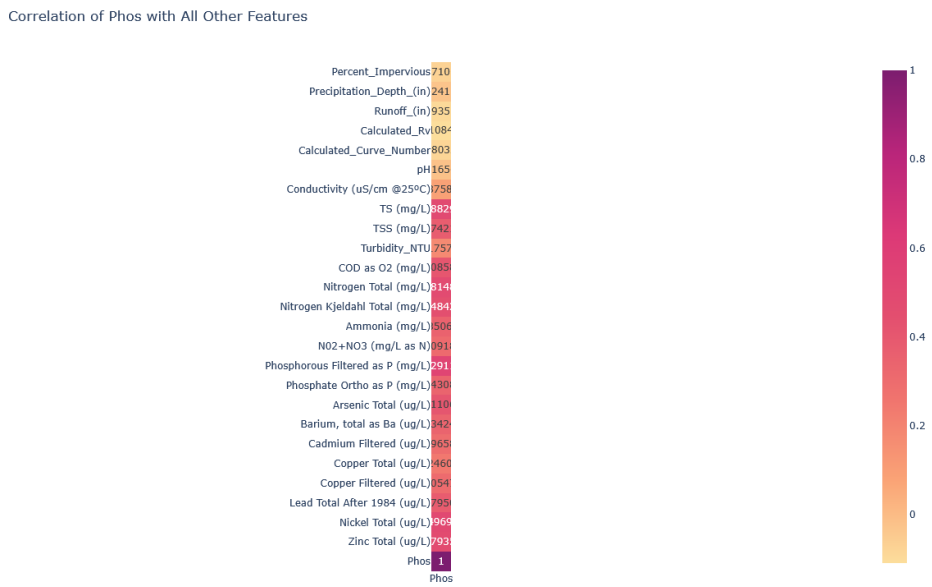

Figure B8. Correlation heatmap for Total Phosphorus.

Figure B9. Correlation heatmap for Total Suspended Solids.